

Supplementary material: Masking cannot remove what is inside the region of interest

S1. Pre-registration index and registry

Every experiment was specified in a dated document committed to the public repository before any model was fitted on the corresponding data; amendments were committed before the amended analysis. The registry lists, for every registration, its first commit and the first commit that added outcome results (outcome-free cell counts excluded; for registrations that add arms to an existing results folder, the first commit whose results contain the new arm). Commit times are set by the committing machine and are therefore not an external time stamp; the registrations of the second review round (`PREREGISTRATION_REVIEW2`) were additionally pushed to GitHub before any model was fitted. The primary family used for multiplicity control (`STATISTICAL_PLAN`) was assembled after the individual results were available; each member test was individually pre-registered, and the Holm correction is applied to the whole family. Generated by `scripts/verify/registry_table.py`.

Registration	Registered (commit, UTC)	First outcome result (commit, UTC)	Order
<code>PREREGISTRATION_ISIC2019_TRAPS</code>	76e25d4 2026-09-26 13:14	2699d51 2026-09-26 15:47	registration first
<code>PRECOMMIT_MATCHED_TRAPS</code>	5ef3228 2026-09-26 14:57	2699d51 2026-09-26 15:47	registration first
<code>PREREGISTRATION_CXR_DEVICE_TRAPS</code>	4d1a924 2026-09-26 18:07	e0d5310 2026-09-26 18:44	registration first
<code>PREREGISTRATION_UNIVERSAL_I2E</code>	ee95f10 2026-09-26 20:14	2f09c07 2026-09-26 21:07	registration first
<code>PREREGISTRATION_THYROID_TRAPS</code>	ea57661 2026-09-26 22:46	b7bc5e7 2026-09-27 00:07	registration first
<code>PREREGISTRATION_SLAS</code>	2d1ba20 2026-09-26 23:33	2d1ba20 2026-09-26 23:33	same commit
<code>PREREGISTRATION_CAPSULE_TRAPS</code>	fcfb8d9 2026-09-27 00:40	6c3e8c5 2026-09-27 00:59	registration first
<code>PREREGISTRATION_CXR_DRAIN</code>	3d0cffc 2026-09-27 03:29	1ef04b7 2026-09-27 04:16	registration first
<code>PREREGISTRATION_FINETUNE</code>	7c8b466 2026-09-27 07:06	1f4f6e3 2026-09-27 08:00	registration first
<code>PREREGISTRATION_OVARY_TRAPS</code>	2517600 2026-09-27 07:29	cd67b8e 2026-09-27 07:45	registration first
<code>STATISTICAL_PLAN</code>	4e9d44c 2026-09-27 09:37	4e9d44c 2026-09-27 09:37	same commit
<code>PREREGISTRATION_NATURAL</code>	74bd243 2026-09-27 09:39	be95223 2026-09-27 09:50	registration first
<code>PREREGISTRATION_INPAINT_LAMA</code>	715b790 2026-09-27 09:41	ae374fd 2026-09-27 10:14	registration first
<code>PREREGISTRATION_TEXT_PROMPT</code>	7877e8c 2026-09-27 12:21	2203aa2 2026-09-27 12:54	registration first
<code>PREREGISTRATION_THEORY_PREDICTION</code>	4f78653 2026-09-27 13:46	30ce70a 2026-09-27 14:03	registration first
<code>PREREGISTRATION_EMBEDDING_GROUPS</code>	b0eda23 2026-09-27 13:55	30ce70a 2026-09-27 14:03	registration first
<code>PREREGISTRATION_SCALE</code>	13303b0 2026-09-27 14:13	cef4871 2026-09-27 14:49	registration first
<code>PREREGISTRATION_UMTE_ABLATION</code>	30f19c8 2026-09-27 14:15	0abd4a7 2026-09-27 14:22	registration first
<code>PREREGISTRATION_FT_ERASE</code>	8ac7a7d 2026-09-27 14:20	3d0dd70 2026-09-27 14:36	registration first
<code>PREREGISTRATION_FT_UMTE</code>	b7d7077 2026-09-27 14:52	f28ab78 2026-09-27 15:27	registration first
<code>PREREGISTRATION_REVIEW2</code>	29cd0da 2026-09-27 18:56	3673625 2026-09-27 19:40	registration first

S2. Replication of the original pilot

Of 88 numbers reported in the thesis proposal, 73 are reproduced exactly (same sign, same confidence-interval verdict, $|\Delta| \leq 0.02$), 12 agree in sign and significance with a larger difference, and 3 differ (a borderline occlusion interval, a contrast trap that was reported as null and is not, and one source stratum). Details: `results/SUMMARY.md`. The archived pilot confidence intervals are recomputed exactly from the archived predictions (`make verify`).

S3. Datasets considered and not used

CANDID-PTX (not accessible outside Health New Zealand); RANZCR-CLiP devices linked to NIH (masking helped at both locations; device placement confounded with other devices); chest-radiograph cardiomegaly with lines over the heart (210 labelled images); TCGA/GrandQC pathology (pen-mark masks unreliable); ISIC ink (70 in-lesion images); breast ultrasound BUSI/BUS-BRA (caliper-free group only $\approx 70\%$ pure after a two-detector design); DDTI thyroid (TI-RADS instead of pathology; 134 images); BKAI-IGH colonoscopy specular highlights (no highlight-free images). *External validation with patient identifiers*. TN-SCUI 2020 (one image per patient, nodule masks) may not be used outside the challenge under its terms. ThyUS2Path [4] (842 patients, 8,508 images, surgical pathology, CC BY 4.0) contains real calipers in several colours but no nodule masks, and our frozen caliper detector responds to elements of its scanner interface (frame lines, depth scales) on a random 25% sample inspected before any label was read. Using it requires a nodule segmentation transferred from TN3K and a caliper detector re-validated on this interface; we did not run it.

S4. Artifact labels: audits, sensitivity and an expert audit kit

Visual audit of 40 random images per cell by the analysis team (not clinicians): thyroid detections 95% on real calipers; caliper-free thyroid images 95–97% without visible calipers; ovary 40/40 detections are overlays, 31/40 calipers; caliper-free ovary images 92% without calipers but $\approx 28\%$ with other overlays. Capsule debris probe: pixel accuracy 0.92, IoU 0.62 against expert masks. Re-running with large-marker (≥ 50 px) and strict-location ($r \geq 0.7$ / $r < 0.05$) definitions (thyroid, ovary) and a strict capsule Trap B leaves every primary conclusion unchanged (SENSITIVITY_ARTIFACT_LABELS). *Expert audit (prepared, not yet carried out)*. `scripts/audit/make_expert_audit_kit.py` draws 60 images per cohort (half flagged by the automatic label), renders each blind (ROI contour only) and with the automatic mask, and writes a rating sheet (artifact present; location relative to the ROI; mask quality); `scripts/audit/score_expert_audit.py` computes Cohen’s κ between two independent raters and the precision and recall of the automatic labels against their consensus, with bootstrap CIs.

S5. Secondary method analyses

- *Patch-token suppression (SLAS)*. A linear probe on DINOv2 patch tokens localises real hair from 5 annotated images (patch AUROC 0.93; 0.97 with 50), but dropping flagged tokens improves reversed AUROC only modestly (+0.02 hair, +0.08 thyroid over ROI pooling).
- *Annotation-free pseudo-groups*. An overlay detector trained on synthetic insertion pairs recognises real calipers (AUROC 0.84) but not hair (0.57); pseudo-group balancing adds +0.04 on thyroid only.
- *Erasure combined with DFR*. Adds ≤ 0.014 over masking plus DFR.
- *Split-validation selector*. Admitting DFR as a candidate (trained and scored on disjoint validation halves) fixes the chest-drain case (0.656 vs 0.538) but makes choices noisier elsewhere; neither selector is uniformly best.

S6. Leakage

Holding cohort, features and protocol fixed, replacing the grouped split with random image-level splits (93 near-duplicate pairs across train/test) inflates the measured shortcut gap by 18% (0% overlap) and 10% (100%) and distorts both masking effects (+0.184 $\rightarrow$ +0.223; $-0.129 \rightarrow -0.088$).

S7. Backbone scale

Replacing DINOv2 ViT-B/14 (86 M parameters) by ViT-S/14 (21 M) or ViT-L/14 (304 M), with everything else fixed (pre-registered), the location crossover is positive in all nine cohort-size combinations (ViT-S / ViT-B / ViT-L: thyroid +0.244 / +0.230 / +0.271; ovary +0.298 / +0.170 / +0.202; capsule +0.370 / +0.368 / +0.408; every CI excludes zero), and protected U-MtE beats masking on thyroid and ovary at every size (ViT-S +0.181 [+0.146, +0.218] and +0.103 [+0.078, +0.128]; ViT-L +0.222 [+0.195, +0.256] and +0.060 [+0.043, +0.077]). Larger models exploit the in-ROI shortcut *more*: the ERM correlated-reversed gap grows from 0.547 to 0.621 (thyroid) and from 0.319 to 0.570 (ovary) between ViT-S and ViT-L, so scale does not remove it. On capsule, disease protection is less complete with ViT-L (-0.040 [$-0.046, -0.035$] versus masking).

S8. Mitigation methods and their results on the traps

The main text summarises the mitigations as a decision guide. This section gives the annotation-free mask-then-erase method (U-MtE), its results on the traps, the baselines and the end-to-end variants. None of these remedies is claimed to improve performance on unaltered test data (main text).

S8.1 Method: mask-then-erase with a generic insertion library

Insertion pairs. Removing an in-ROI artifact requires its pixels; inserting one does not. Given a frozen encoder f and an overlay generator $\mathcal{G}$, every training image x_i yields a counterfactual $x_i^+ = \mathcal{G}(x_i; \xi_i)$ with random parameters ξ_i and random position, half of the elements centred inside the ROI. Because insertion ignores the label, the differences $\Delta_i = f(x_i^+) - f(x_i)$ are independent of Y .

Generic library. $\mathcal{G}$ is one procedural family used unchanged for every modality and artifact: thin curvilinear strokes, thick bands, solid or translucent patches, blobs, dashed lines and small glyphs, with random colour (half greyscale), opacity and width. It contains no ruler, hair, caliper or debris template; it is a prior over “foreign overlays”, not a model of any particular artifact.

Erasure. The insertion eraser removes the span of the differences:

$$\Sigma_\Delta = \frac{1}{n} \sum_i \Delta_i \Delta_i^\top, \quad U_k = \text{top-}k \text{ eigvecs}(\Sigma_\Delta), \quad P(z) = z - U_k U_k^\top (z - \mu), \quad (1)$$

with k chosen *without labels* as the smallest rank whose held-out paired-difference energy $\mathbb{E}\|P\Delta\|^2 / \mathbb{E}\|\Delta\|^2$ falls below 0.10 (fixed before any experiment).

Mask-then-erase (U-MtE). Masking and erasure have complementary reach: masking removes any artifact outside the ROI but none inside; erasure acts on the representation regardless of location. U-MtE therefore computes the pairs *on masked images* ($\Delta_i = f(m(x_i^+)) - f(m(x_i))$) so that the erased subspace targets exactly what masking leaves behind, and fits the head on $P(f(m(x)))$. When image-level artifact labels exist, the head is group-balanced (U-MtE_{bal}).

Disease protection. If the artifact resembles the pathology (intestinal debris versus erosion fibrin), U_k can contain disease evidence. Let W be an orthonormal basis of logistic label directions fitted on artifact-free training images; we erase $U'_k = \text{orth}((I - WW^\top)U_k)$ instead, which guarantees $w^\top P(z) = w^\top z$ for every $w \in \text{span}(W)$ (cf. task-preserving erasure, 3).

Scope. U-MtE is a representation-level intervention: it acts on the features of a frozen encoder, the dominant way in which foundation models are deployed on small medical datasets, and needs only the encoder, the ROI mask and the generator. An end-to-end analogue is evaluated as a secondary check (S8.3).

Tool. The released package exposes insertion, erasure and a patch-token artifact probe for any frozen backbone (ViTs and CNNs) through a two-call API.

Table 1: In-ROI artifacts (Trap A): reversed-test AUROC, with min(reversed, correlated) in parentheses (penalises shortcut flipping). Best robust value per column in bold. Generated by `scripts/make_main_table.py`.

Method	Needs	Hair/DINOv2	Hair/DermLIP	Thyroid/DINOv2	Thyroid/MedSigLIP	Thyroid/ConvNeXt	Ovary/DINOv2	Ovary/MedSigLIP	Capsule/DINOv2	Capsule/MedSigLIP	Capsule/ConvNeXt	Drain/RAD-DINO	Drain/DINOv2
ERM	—	0.49 (0.49)	0.67 (0.67)	0.29 (0.29)	0.28 (0.28)	0.35 (0.35)	0.41 (0.41)	0.42 (0.42)	0.59 (0.59)	0.59 (0.59)	0.55 (0.55)	0.19 (0.19)	0.28 (0.28)
ROI masking	ROI	0.45 (0.45)	0.58 (0.58)	0.40 (0.40)	0.35 (0.35)	0.36 (0.36)	0.55 (0.55)	0.43 (0.43)	0.68 (0.68)	0.70 (0.70)	0.64 (0.64)	0.24 (0.24)	0.29 (0.29)
Mask + overlay augmentation	ROI	0.46 (0.46)	—	0.42 (0.42)	0.38 (0.38)	0.37 (0.37)	0.53 (0.53)	—	0.70 (0.70)	0.72 (0.72)	0.65 (0.65)	0.24 (0.24)	0.29 (0.29)
U-MtE (ours)	ROI	0.51 (0.51)	0.51 (0.51)	0.62 (0.62)	0.55 (0.55)	0.49 (0.49)	0.56 (0.56)	0.64 (0.64)	0.57 (0.57)	0.47 (0.47)	0.56 (0.56)	0.27 (0.27)	0.28 (0.28)
U-MtE, protected (ours)	ROI + Y of A=0	0.49 (0.49)	0.59 (0.59)	0.62 (0.62)	0.61 (0.61)	0.55 (0.55)	0.59 (0.59)	0.57 (0.57)	0.67 (0.67)	0.67 (0.67)	0.64 (0.64)	0.27 (0.27)	0.28 (0.28)
Group-balanced	A	0.72 (0.72)	0.81 (0.81)	0.68 (0.68)	0.70 (0.70)	0.66 (0.66)	0.71 (0.71)	0.73 (0.73)	0.78 (0.78)	0.81 (0.81)	0.72 (0.72)	0.54 (0.54)	0.40 (0.40)
DFR	A	0.74 (0.74)	0.78 (0.78)	0.74 (0.66)	0.76 (0.72)	0.71 (0.67)	0.72 (0.69)	—	0.80 (0.80)	0.80 (0.80)	0.77 (0.77)	0.68 (0.68)	0.55 (0.55)
Mask + DFR	ROI + A	0.69 (0.69)	—	0.83 (0.74)	0.81 (0.76)	0.77 (0.72)	0.75 (0.75)	—	0.88 (0.88)	0.90 (0.90)	0.86 (0.86)	0.65 (0.65)	0.57 (0.57)
U-MtE + balanced (ours)	ROI + A	0.70 (0.70)	0.65 (0.65)	0.79 (0.79)	0.79 (0.79)	0.74 (0.74)	0.74 (0.74)	0.79 (0.79)	0.86 (0.86)	0.85 (0.85)	0.80 (0.80)	0.46 (0.46)	0.44 (0.44)

S8.2 Results on the traps

U-MtE repairs what masking leaves behind. Without any artifact example, artifact mask or artifact label—it uses only the ROI mask that masking already requires—U-MtE improves reversed AUROC over masking in the in-ROI regime of every modality: thyroid calipers $+0.222$ $[+0.197, +0.251]$ (DINOv2) and $+0.204$ $[+0.174, +0.237]$ (MedSigLIP); dermoscopic hair $+0.052$ $[+0.044, +0.062]$; synthetic capsule debris $+0.081$ $[+0.050, +0.112]$; synthetic calipers $+0.278$ $[+0.232, +0.325]$; synthetic chest tubes $+0.134$ $[+0.120, +0.151]$ (DINOv2 / RAD-DINO). With MedSigLIP the unbalanced method *loses* to masking in both controlled sweeps (debris -0.114 $[-0.147, -0.082]$, calipers -0.076 $[-0.110, -0.043]$) although it wins on real thyroid calipers ($+0.204$); the balanced variant beats masking in all four controlled sweeps ($+0.193$ to $+0.471$) and has no location interaction in any of them. Disease protection turns the MedSigLIP debris failure into a gain ($+0.061$ $[+0.032, +0.092]$ over masking) but not the caliper one (-0.041 $[-0.089, -0.005]$).

Sensitivity to the erasure rank. In a pre-registered ablation (Trap A, DINOv2), the thyroid gain over masking is $+0.18$ to $+0.22$ for every setting that erases at least 16 directions (held-out energy 0.5–0.99, or fixed $k \in \{16, 64\}$), but only $+0.06$ to $+0.07$ with one to four directions: the shift caused by overlays is multi-dimensional. The pre-specified 90% rule reached the rank cap ($k = 64$) in all three cohorts. The protected variant never harms capsule by more than 0.010 and helps ovary at every rank ($+0.007$ to $+0.043$), whereas unprotected erasure harms capsule more the more directions it removes, as the overlap term ρ of the linear-Gaussian account (main paper) predicts.

Erasure, not augmentation. Using the same generic overlays as training augmentation (overlaid copies of a random half of the training images, in the masked view) gains at most $+0.03$ over masking; erasing their subspace instead gains $+0.207$ $[+0.182, +0.235]$ (thyroid, DINOv2), $+0.177$ $[+0.148, +0.211]$ (thyroid, MedSigLIP) and $+0.045$ $[+0.037, +0.054]$ (hair) more than augmentation. Under the stricter near-duplicate grouping (main paper, stricter leakage grouping) the thyroid advantage is unchanged ($+0.192$) but the small ovary advantage ($+0.035$) loses significance, so we regard erase $>$ augment as established for thyroid and hair only.

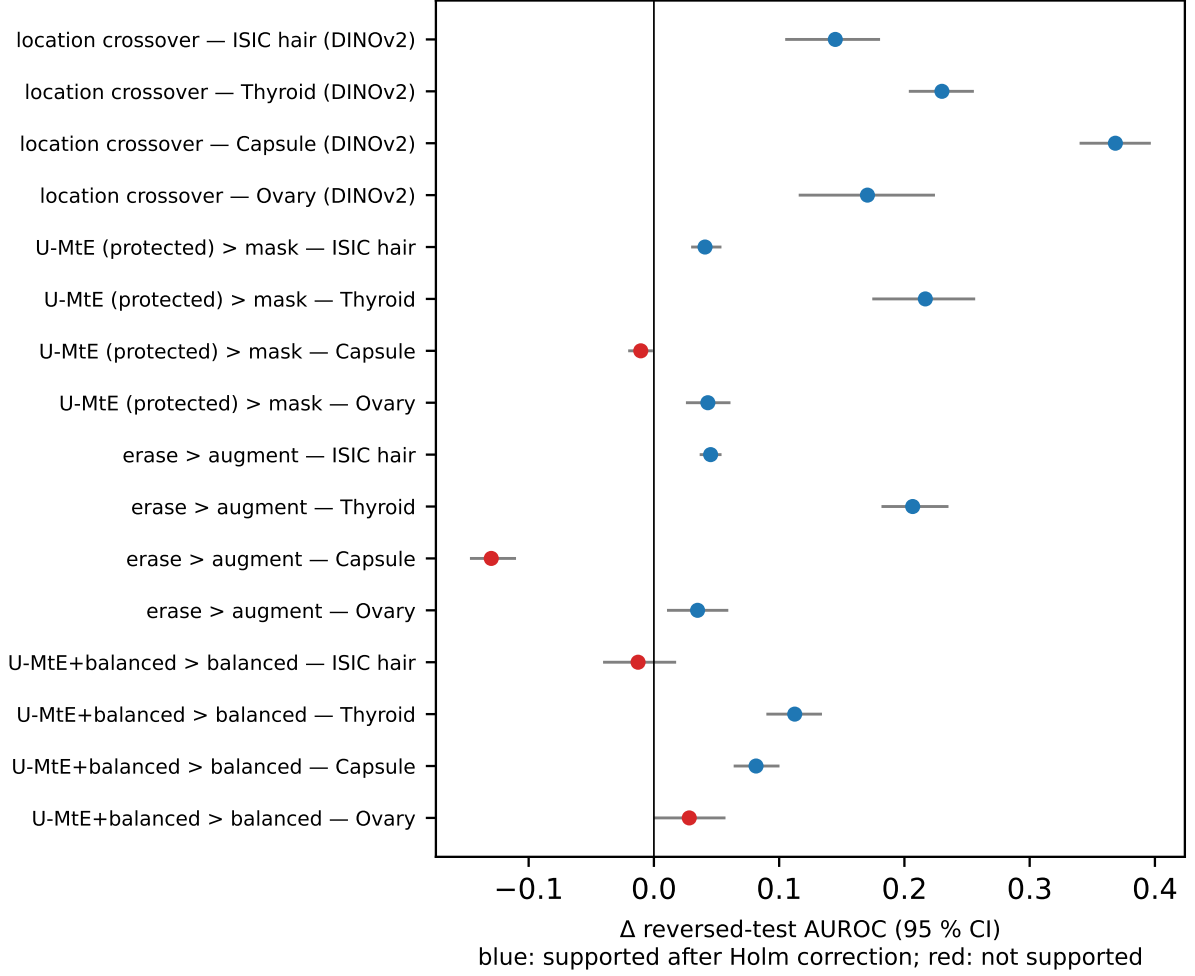


Figure 1: The 16 pre-registered primary tests with Holm correction (blue: supported; red: not supported). Generated by `scripts/analysis/primary_claims.py` and `scripts/make_figures.py`.

When the artifact resembles the disease. On real capsule debris U-MtE is worse than masking (-0.115): erosions carry yellow-white fibrin that resembles debris, and a subspace estimated from real-debris templates is worse still (-0.395). Disease protection removes the failure (-0.010 [$-0.020, -0.001$] generic, -0.012 [$-0.028, +0.003$] template) at a cost of at most 0.012 where erasure already works (thyroid -0.005 [$-0.036, +0.024$], hair -0.012 [$-0.020, -0.003$]).

With image-level artifact labels. Adding group balancing, U-MtE_{bal} is the most robust model in thyroid ultrasound by the minimum of correlated and reversed AUROC (0.788 DINOv2, 0.792 MedSigLIP; DFR 0.659 and 0.719); in real capsule endoscopy masking plus last-layer retraining is higher (0.885 versus 0.857); on dermoscopic hair DFR alone is best (0.741) because most of the hair shortcut is not in the hair pix-

els (removing every hair patch with the oracle mask gains only +0.015). In both controlled sweeps the benefit of U-MtE_{bal} no longer depends on location (interaction -0.000 $[-0.023, +0.023]$ capsule; -0.054 $[-0.098, -0.004]$ thyroid), whereas that of masking does (+0.205, +0.432). The capsule findings replicate with MedSigLIP (crossover +0.332 $[+0.296, +0.365]$; masking plus DFR 0.901). All thyroid and capsule findings replicate with a CNN (ConvNeXt-Base): crossovers +0.284 and +0.292; U-MtE – mask +0.128 $[+0.112, +0.145]$ and erase – augment +0.120 $[+0.102, +0.139]$ on thyroid.

Label-free and erasure baselines. Among methods that need no artifact labels, U-MtE beats JTT [5] by +0.198 $[+0.167, +0.231]$ (thyroid), +0.137 $[+0.098, +0.179]$ (ovary) and +0.094 $[+0.077, +0.109]$ (chest drains, RAD-DINO; not with DINOv2, -0.010); JTT on the unmasked view is best on hair. Task-preserving concept removal (SPLINCE; 3), fitted with artifact labels in the whitened, minimal-distortion form of the original method, removes the artifact covariance at a large cost in disease signal (clean AUROC 0.56–0.78 versus 0.69–0.87 for masking) and still flips the shortcut on capsule endoscopy (reversed 0.868, correlated 0.683): when artifact and label are strongly associated in training, removing one removes much of the other. By $\min(\text{reversed}, \text{correlated})$ U-MtE_{bal} exceeds SPLINCE on thyroid, capsule and hair (0.788 vs 0.596; 0.857 vs 0.683; 0.704 vs 0.564); on chest drains SPLINCE is higher (0.609 vs 0.460) but below DFR (0.683).

Text-prompted erasure. With vision–language backbones an artifact direction can be obtained from text alone (“a thyroid ultrasound image with measurement calipers” minus “a thyroid ultrasound image”; biased-prompt construction, 1). Erasing that subspace after masking is a near no-op in every cohort (+0.001 to +0.008 over masking; MedSigLIP thyroid, ovary, capsule and DermLIP hair), whereas U-MtE with the same MedSigLIP features gains +0.204 $[+0.174, +0.237]$ (thyroid) and +0.215 $[+0.182, +0.246]$ (ovary). The direction along which a described artifact moves the *text* embedding is not the direction along which the real artifact moves the *image* embedding; synthetic insertions measure the latter directly.

A single procedure: validation-driven selection. Selecting, per fold, the candidate with the best group-aware validation score (image-level artifact labels needed on validation images only) beats masking in all five cohorts (+0.17 to +0.40), is within 0.02 of the best candidate in each, and beats DFR in three (thyroid +0.057, capsule +0.047, ovary +0.010); it chooses U-MtE_{bal} where the artifact is an overlay and plain balancing on hair and drains, where DFR remains better. A split-validation variant that admits DFR as a candidate is reported in the supplement.

Modern inpainting. Inpainting the detected caliper pixels with LaMa [6] and then masking is a strong baseline when artifact masks are available (thyroid +0.226 $[+0.213, +0.238]$, ovary +0.123 $[+0.089, +0.155]$ over masking). U-MtE_{protect}, which needs no artifact mask, matches it on thyroid (-0.009 $[-0.045, +0.023]$) and is below it on ovary (-0.080) and on dermoscopic hair, where published hair masks are available (-0.086); with image-level artifact labels, U-MtE_{bal} exceeds it in all three (+0.162, +0.067, +0.123).

S8.3 End-to-end fine-tuning and erase-while-fine-tuning

Fine-tuning ResNet-50 end-to-end (five folds) reproduces the frozen-backbone findings. Capsule: masking harms in-ROI ($-0.284 [-0.347, -0.226]$) and helps out-of-ROI ($+0.301$), crossover $+0.584 [+0.469, +0.691]$. Thyroid: the crossover points the same way ($+0.057 [-0.021, +0.130]$, five folds), and an end-to-end analogue of U-MtE—an invariance penalty between masked images with and without a generic overlay—improves on masking by $+0.143 [+0.099, +0.188]$ without shortcut flipping (reversed 0.679, correlated 0.703), matching the best label-using arm. It fails on capsule (-0.021), where the artifact resembles the pathology. Ovary (1,202 images) is underpowered: U-MtE end-to-end has the best reversed, correlated and clean AUROC, with CIs overlapping masking. A fine-tuned ViT-S reproduces the crossover on thyroid ($+0.168 [+0.074, +0.265]$), but the end-to-end U-MtE analogue does not improve on masking with this architecture ($+0.009 [-0.021, +0.040]$). Applying U-MtE post hoc to the penultimate features of the masked, fine-tuned ResNet-50 does not help either (pre-registered; $+0.004 [+0.000, +0.012]$ over the same features without erasure): after task training the shortcut is encoded along task-specific directions that generic overlays do not move. For fully fine-tuned networks the subspace must be estimated and removed *during* training. *Erase-while-fine-tuning* (pre-registered; no artifact annotation) projects a running estimate of the overlay-shift subspace of the network’s own penultimate features out before the head at every step and at test time, and adds feature invariance and prediction consistency between masked images with and without a generic overlay. On thyroid it improves on masking by $+0.157 [+0.114, +0.205]$ with ResNet-50 and $+0.143 [+0.103, +0.187]$ with ViT-S (reversed AUROC 0.693 and 0.608 vs 0.536 and 0.465), whereas each component alone works for only one architecture (projection + invariance: ResNet-50 $+0.160 [+0.114, +0.207]$, ViT-S $+0.061 [-0.005, +0.141]$; prediction consistency: ResNet-50 $+0.007 [-0.014, +0.029]$, ViT-S $+0.062 [+0.031, +0.087]$). With ResNet-50 the correlated AUROC falls slightly below the reversed one (0.661 vs 0.693), a mild over-correction. On ovary (ResNet-50) it is positive but not significant ($+0.094 [-0.011, +0.210]$; every fine-tuning contrast on this 1,202-image cohort is underpowered), and on capsule, where debris resembles erosion fibrin, it fails ($-0.022 [-0.046, +0.001]$), as does every annotation-free method.

S9. Operating points on every unaltered test set

Masking minus ERM at the five pre-registered validation-fixed operating points (95% hierarchical bootstrap CIs; thresholds re-estimated in every replicate). *Conflicting*: positive cases whose in-ROI artifact status points to the negative class. Generated by `scripts/analysis/operating_points.py`.

Test set	Threshold (validation)	Δ sensitivity	Δ sensitivity, conflicting	Δ specificity
Thyroid	maxBA	-0.275 [-0.399, -0.182]	-0.479 [-0.583, -0.362]	+0.202 [+0.140, +0.300]
	spec0.80	-0.202 [-0.265, -0.134]	-0.372 [-0.436, -0.291]	+0.117 [+0.082, +0.151]
	spec0.90	-0.152 [-0.225, -0.071]	-0.236 [-0.345, -0.132]	+0.075 [+0.043, +0.109]
	sens0.80	-0.373 [-0.431, -0.310]	-0.546 [-0.614, -0.480]	+0.278 [+0.229, +0.329]
	sens0.90	-0.281 [-0.353, -0.221]	-0.462 [-0.560, -0.369]	+0.339 [+0.269, +0.396]
ISIC, BCN held out	maxBA	+0.000 [-0.030, +0.041]	+0.001 [-0.032, +0.044]	-0.078 [-0.152, -0.014]
	spec0.80	+0.017 [+0.003, +0.030]	+0.019 [+0.003, +0.033]	-0.110 [-0.130, -0.086]
	spec0.90	+0.042 [+0.018, +0.065]	+0.042 [+0.013, +0.070]	-0.104 [-0.131, -0.077]
	sens0.80	+0.002 [-0.024, +0.028]	+0.003 [-0.023, +0.030]	-0.082 [-0.130, -0.029]
	sens0.90	+0.001 [-0.021, +0.022]	-0.001 [-0.027, +0.023]	-0.065 [-0.120, -0.005]
ISIC, HAM held out	maxBA	-0.006 [-0.085, +0.113]	+0.002 [-0.078, +0.125]	+0.042 [-0.043, +0.102]
	spec0.80	-0.062 [-0.103, -0.025]	-0.055 [-0.095, -0.018]	+0.092 [+0.068, +0.120]
	spec0.90	-0.057 [-0.110, -0.015]	-0.053 [-0.107, -0.007]	+0.041 [+0.024, +0.066]
	sens0.80	-0.004 [-0.040, +0.032]	+0.009 [-0.033, +0.047]	+0.061 [+0.029, +0.101]
	sens0.90	-0.021 [-0.047, +0.007]	-0.013 [-0.042, +0.020]	+0.096 [+0.053, +0.146]
ISIC, MSK held out	maxBA	-0.013 [-0.085, +0.054]	-0.033 [-0.105, +0.035]	+0.032 [-0.021, +0.086]
	spec0.80	-0.074 [-0.107, -0.046]	-0.094 [-0.128, -0.065]	+0.083 [+0.072, +0.100]
	spec0.90	-0.091 [-0.129, -0.060]	-0.110 [-0.151, -0.077]	+0.058 [+0.047, +0.070]
	sens0.80	-0.050 [-0.086, -0.017]	-0.070 [-0.107, -0.037]	+0.068 [+0.046, +0.095]
	sens0.90	-0.023 [-0.065, +0.017]	-0.037 [-0.081, +0.005]	+0.073 [+0.032, +0.112]
Capsule	maxBA	+0.086 [+0.033, +0.150]	-0.074 [-0.141, +0.012]	+0.089 [+0.040, +0.142]
	spec0.80	+0.140 [+0.086, +0.211]	+0.022 [-0.025, +0.106]	+0.041 [-0.023, +0.102]
	spec0.90	+0.245 [+0.171, +0.307]	+0.092 [-0.017, +0.195]	-0.000 [-0.035, +0.053]
	sens0.80	+0.030 [-0.009, +0.069]	-0.096 [-0.173, -0.029]	+0.092 [+0.048, +0.148]
	sens0.90	+0.022 [-0.020, +0.062]	-0.104 [-0.178, -0.048]	+0.169 [+0.097, +0.268]

S10. Absolute performance and published results

Only the thyroid cohort has a published benchmark on the same split and labels (official patient-disjoint TNCD split, 614 test images). The ovarian (benign cystic versus solid-component tumours) and capsule (erosion versus polyp-like) binary tasks and the ISIC 2019 leave-one-hospital-out melanoma task were defined for this study and have no published counterpart; their clean AUROC is reported with every result table. Frozen encoders with a linear head are below fine-tuned CNNs; the location law replicates with fine-tuned networks (main text).

Model	Source	Test AUROC
ResNet-18, cross-entropy (fine-tuned)	Gong et al. [2]	0.773
DenseNet-121, cross-entropy (fine-tuned)	Gong et al. [2]	0.780
ConvNeXt-T, cross-entropy (fine-tuned)	Gong et al. [2]	0.784
ResNet-18, adaptive curriculum (fine-tuned)	Gong et al. [2]	0.799
DINOv2 ViT-B/14 frozen + linear head, ERM (archived run)	this work	0.736
DINOv2 ViT-B/14 frozen + linear head, ERM (rebuilt run)	this work	0.735

Table 2: Covariate balance of the location traps: standardised mean differences (SMD) between the artifact-bearing images of Trap A (in-ROI) and Trap B (out-of-ROI), before and after propensity matching (categorical covariates: largest level). Positive: larger in Trap A.

Cohort	Covariate	SMD before	SMD after
ISIC hair	lesion area	+2.13	+0.03
	hair amount	-0.16	+0.11
	age	+0.14	-0.01
	age missing	-0.04	-0.05
	sex (missing)	-0.03	-0.04
	site (torso)	+0.19	-0.01
	benign dx (MEL)	+0.26	+0.00
	source (MSK)	-0.54	+0.00
Thyroid	roi area	+0.40	-0.01
	caliper px log	+0.53	+0.04
	width	-0.14	+0.04
	height	-0.12	+0.02
	mean grey	+0.18	-0.01
	roi centroid row	+0.06	+0.01
Ovary	roi area	+0.32	-0.00
	caliper px log	+0.39	+0.03
	width	-0.30	+0.04
	height	+0.05	+0.00
	mean grey	+0.11	+0.02
	subtype (2.0)	-0.16	-0.01
Capsule	roi area	+2.57	+0.01
	contamination	-0.33	-0.12
	mean grey	+0.13	+0.11
	frame block (0.0)	-0.37	-0.23

S11. Covariate balance of the traps and the real-artifact transplant

Transplant design. Recipients: artifact-free images of each cohort. Donors: artifact-bearing images of the same cohort; an artifact instance is the largest connected component of its mask after a grouping dilation (calipers 9 px, hair and debris 5 px), with at least 30 pixels and a bounding-box side of at most 259 px. For every recipient, donors are tried in a stable random order (up to $50 \text{ donors} \times 8 \text{ dihedral transforms}$) until one fits both inside ($\text{overlap} \geq 0.95$) and outside ($\text{overlap} = 0$) the ROI within the field of view; the instance is alpha-composited with a Gaussian-blurred mask. Environments as in the controlled sweeps (0.9/0.1 training and correlated test, 0.1/0.9 reversed test, clean test without artifact), 5-fold grouped cross-validation, five presence seeds (bootstrap clusters). Recipients without a feasible donor (mostly small capsule lesion boxes) are excluded; counts by label are in `results/review2/transplant/*/feasibility_by_label.csv`.

Table 3: Real-artifact transplant: reversed / correlated / clean AUROC of ERM and ROI masking with the same real artifact inside or outside the ROI of the same images, and the change from masking (95% CI). $|\Delta p|$: same-head counterfactual sensitivity to inserting the artifact.

Cohort	Location	ERM rev / corr / clean	Mask rev / corr / clean	Mask – ERM (rev)	$ \Delta p $ ERM / mask
ISIC hair	outside ROI	0.539 / 0.869 / 0.759	0.716 / 0.716 / 0.716	+0.178 [+0.154, +0.203]	0.230 / 0.000
	inside ROI	0.454 / 0.904 / 0.776	0.379 / 0.842 / 0.672	−0.075 [−0.095, −0.055]	0.340 / 0.297
Thyroid	outside ROI	0.497 / 0.857 / 0.708	0.846 / 0.846 / 0.846	+0.349 [+0.334, +0.365]	0.201 / 0.000
	inside ROI	0.487 / 0.851 / 0.708	0.559 / 0.906 / 0.801	+0.072 [+0.058, +0.086]	0.206 / 0.271
Ovary	outside ROI	0.686 / 0.759 / 0.725	0.750 / 0.750 / 0.750	+0.064 [+0.035, +0.092]	0.023 / 0.000
	inside ROI	0.657 / 0.771 / 0.723	0.647 / 0.825 / 0.755	−0.010 [−0.041, +0.022]	0.037 / 0.116
Capsule	outside ROI	0.592 / 0.905 / 0.795	0.914 / 0.914 / 0.914	+0.322 [+0.289, +0.356]	0.193 / 0.000
	inside ROI	0.508 / 0.906 / 0.787	0.333 / 0.956 / 0.871	−0.175 [−0.210, −0.141]	0.238 / 0.551

S12. CLAIM 2024 checklist

Checklist for Artificial Intelligence in Medical Imaging, 2024 update [7]. M = main paper section, S = supplement section.

No.	Item	Answer	Where / comment
<i>Title / abstract</i>			
1	Identification as a study of AI methodology, specifying the category of technology used (eg, deep learning)	Yes	Title, abstract: deep-learning classifiers on frozen foundation-model encoders and fine-tuned CNN/ViT.
2	Summary of study design, methods, results, and conclusions	Yes	Abstract.
<i>Introduction</i>			
3	Scientific and/or clinical background, including the intended use and role of the AI approach	Yes	M Introduction, Related work.
4	Study aims, objectives, and hypotheses	Yes	M Introduction; hypotheses pre-registered (S1).
<i>Methods — study design</i>			
5	Prospective or retrospective study	Yes	Retrospective analysis of public datasets (M Data).
6	Study goal	Yes	Methodological study of shortcut mitigation (not a diagnostic-accuracy study); M Introduction, Setup.
<i>Methods — data</i>			
7	Data sources	Yes	M Data (ISIC 2018/2019, HAM10000 masks, TN3K/TNCD, MMOTU, SEE-AI, NIH ChestX-ray14); licences in docs/DATA.md.
8	Inclusion and exclusion criteria	Yes	M Data (trap cells, artifact-free groups, QC); transplant feasibility (S11).
9	Data preprocessing	Yes	M Setup (518px rendering, ROI masking, feature caching).
10	Selection of data subsets	Yes	Traps, matched traps and transplant cohorts (M Results, S11).
11	De-identification methods	NA	Public datasets released de-identified by their providers.
12	How missing data were handled	Yes	Missing age imputed with an indicator; images without a lesion mask excluded by QC (152 ISIC images).

No.	Item	Answer	Where / comment
13	Image acquisition protocol	Partial	As documented by the dataset providers (cited); no acquisition parameters are released for TN3K, MMOTU or SEE-AI.
<i>Methods — reference standard</i>			
14	Definition of method(s) used to obtain reference standard	Yes	Diagnoses as released by the dataset providers (TNCD: cytological biopsy; ISIC: histopathology, follow-up or expert consensus; MMOTU tumour categories; SEE-AI expert lesion annotation) (M Data).
15	Rationale for choosing the reference standard	Yes	Dataset reference standards; ROI masks from manual annotation or a U-Net (M Data).
16	Source of reference standard annotations	Yes	Dataset annotators; artifact labels from a rule-based detector, a patch probe and published masks (M Data).
17	Annotation of test set	Yes	Same procedures as for training images.
18	Measures of inter- and intrarater variability of features described by the annotators	No	Artifact labels were audited by the analysis team only; a blinded two-rater expert audit kit with Cohen’s κ is prepared (S4) but not yet carried out (M Limitations).
<i>Methods — data partitions</i>			
19	How data were assigned to partitions	Yes	Five seeds of five-fold group-safe cross-validation; official TNCD split; leave-one-hospital-out (M Setup, Consequences on unaltered data).
20	Level at which partitions are disjoint	Yes	Lesion identifiers (ISIC), patient level for the official TNCD split and chest radiographs, otherwise perceptual-hash and frame-block groups; embedding-group sensitivity (M Robustness).
<i>Methods — testing data</i>			
21	Intended sample size	No	No a-priori sample-size calculation; all eligible images used; underpowered contrasts are identified (ovary).
<i>Methods — model</i>			
22	Detailed description of model	Yes	Frozen DINOv2 ViT-S/B/L, MedSigLIP, DermLIP, RAD-DINO, ConvNeXt with logistic heads; fine-tuned ResNet-50 and ViT-S (M Setup, Replication; S8).
23	Software libraries, frameworks, and packages	Yes	<code>requirements.txt</code> (PyTorch 2.11, timm, scikit-learn, open_clip) (S13).
24	Initialization of model parameters	Yes	Published pre-trained checkpoints; ImageNet initialisation for fine-tuning.
<i>Methods — training</i>			
25	Details of training approach	Yes	Regularisation and decision threshold on clean validation data; fine-tuning recipe (S8).
26	Method of selecting the final model	Yes	Clean-validation AUROC; no test-set selection.
27	Ensembling techniques	NA	No ensembles.
<i>Methods — evaluation</i>			
28	Metrics of model performance	Yes	Reversed, correlated and clean AUROC; location crossover; AUPRC, Brier, ECE, sensitivity/specificity (M Setup, Consequences on unaltered data).
29	Statistical measures of significance and uncertainty	Yes	Hierarchical paired bootstrap (10,000); Holm-corrected primary family (M Setup; S1).
30	Robustness or sensitivity analysis	Yes	Label definitions, leakage grouping, covariate matching, rebuild on new hardware (M Robustness).
31	Methods for explainability or interpretability	Yes	Same-head counterfactual sensitivity to inserting the artifact (M Mechanism).
32	Evaluation on internal data	Yes	All traps and sweeps.

No.	Item	Answer	Where / comment
33	Testing on external data	Partial	ISIC leave-one-hospital-out; no external cohort for ultrasound or capsule endoscopy (M Limitations; S3).
34	Clinical trial registration	NA	Retrospective; analysis pre-registrations in the repository (S1).
<i>Results</i>			
35	Numbers of patients or examinations included and excluded	Yes	Image counts per cohort, trap cell and transplant feasibility; patients where identifiers exist (M Data; S11).
36	Demographic and clinical characteristics of cases in each partition	Partial	Age, sex, body site and diagnosis for ISIC; lesion size and acquisition covariates for all cohorts (S11); demographics are not released for TN3K, MMOTU or SEE-AI.
37	Performance metrics and measures of statistical uncertainty	Yes	All result tables give 95% CIs.
38	Estimates of diagnostic performance and their precision	Yes	Sensitivity and specificity at five pre-registered operating points with CIs (M Consequences on unaltered data; S9).
39	Failure analysis of incorrectly classified cases	Yes	Shortcut-conflicting cases; DermLIP without a hair gap; chest radiography boundary case (M Results).
<i>Discussion</i>			
40	Study limitations	Yes	M Discussion and limitations.
41	Implications for practice, including intended use and/or clinical role	Yes	M Discussion; decision guide (M What carries the shortcut decides the fix).
<i>Other information</i>			
42	Provide a reference to the full study protocol or to additional technical details	Yes	Pre-registration documents and registry (S1); report/full/full_report.pdf .
43	Statement about the availability of software, trained model, and/or data	Yes	Code, pre-registrations and result files are public; all datasets are public (M Conclusion).
44	Sources of funding and other support; role of funders	—	To be completed by the authors.

S13. Reproducibility

All numbers in the main paper and this supplement are produced by scripts in the repository (**scripts/**); tables are generated, never typed, and every confidence interval in the paper is traced automatically to a result file (**scripts/verify/audit_paper_numbers.py**). A complete rebuild on different hardware reproduces every primary crossover within 0.01 (main text). Compute: one 24 GB GPU.

References

- [1] Chuang, C.Y., Jampani, V., Li, Y., Torralba, A., Jegelka, S., 2023. Debiasing vision-language models via biased prompts. arXiv preprint arXiv:2302.00070 .
- [2] Gong, H., Cheng, H., Xie, Y., Tan, S., Chen, G., Chen, F., Li, G., 2022. Less is more: Adaptive curriculum learning for thyroid nodule diagnosis, in: Medical Image Computing and Computer Assisted Intervention (MICCAI). doi:10.1007/978-3-031-16440-8_24.
- [3] Holstege, F., Ravfogel, S., Wouters, B., 2025. Preserving task-relevant information under linear concept removal, in: Advances in Neural Information Processing Systems (NeurIPS). ArXiv:2506.10703.

- [4] Hou, X.W., et al., 2024. An ultrasonography of thyroid nodules dataset with pathological diagnosis annotation for deep learning. *Scientific Data* 11, 1272. doi:10.1038/s41597-024-04156-5.
- [5] Liu, E.Z., Haghighi, B., Chen, A.S., Raghunathan, A., Koh, P.W., Sagawa, S., Liang, P., Finn, C., 2021. Just train twice: Improving group robustness without training group information, in: ICML.
- [6] Suvorov, R., Logacheva, E., Mashikhin, A., Remizova, A., Ashukha, A., Silvestrov, A., Kong, N., Goka, H., Park, K., Lempitsky, V., 2022. Resolution-robust large mask inpainting with fourier convolutions, in: WACV.
- [7] Tejani, A.S., Klontzas, M.E., Gatti, A.A., et al., 2024. Checklist for artificial intelligence in medical imaging (CLAIM): 2024 update. *Radiology: Artificial Intelligence* 6, e240300. doi:10.1148/ryai.240300.

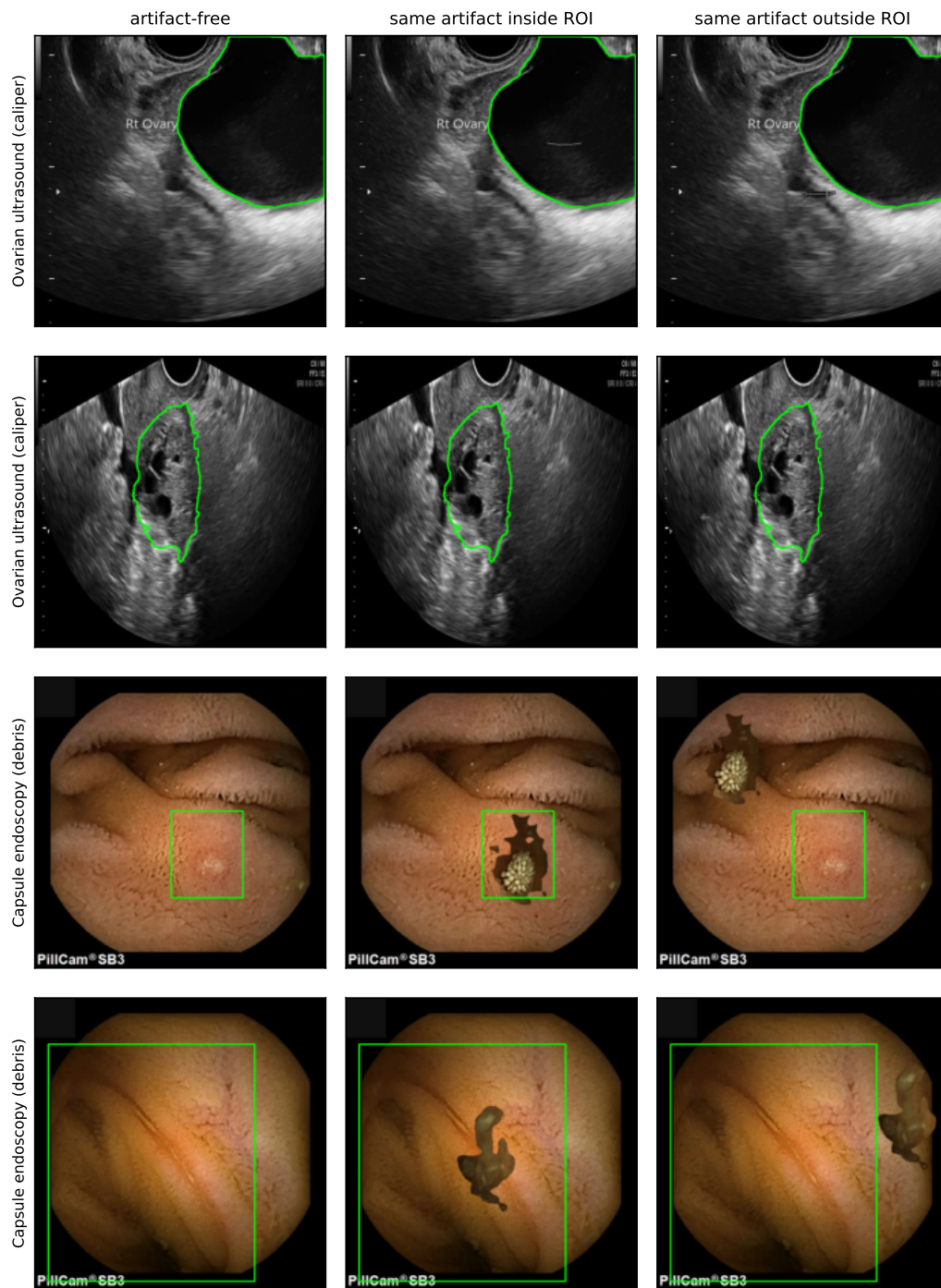


Figure 2: Real-artifact transplant, openly licensed cohorts (green: ROI). Each row shows one artifact-free image and the same real artifact instance—cut from another image of the cohort with its mask—pasted inside and outside the ROI. Instances carry a thin rim of donor background where the mask extends beyond the artifact; the rim is identical at both locations. Generated by `scripts/make_transplant_figure.py`.